# The Triadic Theory of Life and Evolution: The Transgenerational Triadic Cycle, the Unified Triadic Theory of Evolution, and the Mathematics of Heritability and Extinction

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**Abstract**

Biology is electron transfer at scale. We apply the Triadic Redox Theory (TRT) to the full arc of life—states, sex, reproduction, and evolution—as a single continuous theory. Life is the maintenance of Fluidity inside the Operational Window

$$0.5 < \Phi < 2.0, \qquad \Phi \approx 1.6 \text{ optimal for active metabolism,} \tag{1}$$

and death is its irreversible exit. A living system exists in one of four states—**Active**, **Dormant**, **Crystallized**, or **Dead**—distinguished by the electron flow $I$ and the Reinitiation Capacity $\Psi = \max(\Delta G_{\text{restore}})$; a system is alive iff $\Psi > 0$, and entropic death obeys $d\Psi/dt = -k_{\text{entropic}}\Psi$. Reproduction is the **Transgenerational Triadic Cycle (TTC)**: H-like (father) + He-like (mother) $\rightarrow$ Middle Fluid (conception) $\rightarrow$ Crystallized Fluid (offspring) $\rightarrow$ re-liquefaction (puberty) $\rightarrow$ new H-like/He-like adult. We formalize the TTC as a **delay differential equation** for the generational transmission of Fluidity,

$$\frac{d\Phi(t)}{dt} = -\lambda\Phi(t) + \beta\Phi(t-\tau) + \nu\Phi_{\text{env}}(t) + \mu\sin\left(\frac{2\pi t}{\tau_{\text{dev}}}\right) + \xi(t), \tag{2}$$

and derive closed-form results: the steady-state generational Fluidity $\Phi_{\text{steady}} = \nu\Phi_0/(\lambda - \beta e^{-\lambda\tau})$; the **critical heritability threshold**

$$\beta_{\text{crit}} = \lambda e^{\lambda\tau}, \tag{3}$$

which bounds heritability (for $\lambda = 0.05\ \text{yr}^{-1}$, $\tau = 20$ yr: $\beta_{\text{crit}} = 0.136$, so heritability must stay under 13.6%; consistent with the observed human lifespan heritability of $\approx 25\%$ within the $< 30\%$ bound); the optimal generation time $\tau_{\text{opt}} = (1/\lambda)\ln(\beta/\lambda)$; and the decay of transgenerational memory with half-life $t_{1/2} = \ln 2/(\lambda - \beta e^{-\lambda\tau})$. Building on the TTC we derive the **Unified Triadic Theory of Evolution (UTTE)** from a single master equation of gradient ascent on the fitness landscape $W(\boldsymbol{\theta}) = \int_0^\infty \mathbb{P}(0.5 < \Phi(t) < 2.0)\, dt$, unifying Darwinian selection, Mendelian inheritance, and epigenetic transmission. We derive optimal heritability, generation time, aging rate, plasticity, and mutation rate, and the five laws of evolution. The framework yields quantitative predictions (including a $< 30\%$ lifespan-heritability bound and boom–bust population oscillations near the heritability threshold) that extend from the electron to the ecosystem.

**Keywords:** Triadic Redox Theory, Fluidity, Operational Window, Reinitiation Capacity, entropic death, Transgenerational Triadic Cycle, delay differential equation, heritability, Unified Triadic Theory of Evolution, fitness landscape, natural selection, punctuation.

## 1 Introduction

Evolution is the central organizing principle of biology, yet its drivers—mutation, selection, drift, and inheritance—are usually described statistically, without a physical mechanism [1, 2, 3]. Why do organisms age, reproduce, and die at characteristic timescales? Why is lifespan heritability surpisingly low? Why does evolution accelerate in changing environments but stall in stable ones?

The answer proposed here is that all of these phenomena are Redox phenomena. TRT [5] gives a single dimensionless state variable—the Fluidity $\Phi$—that governs every electron-transfer event, and biology is nothing other than the controlled maintenance and transmission of $\Phi$. Section 2 reduces life itself to four states of a redox system. Section 3 treats reproduction as a triadic generational cycle. Section 4 formalizes the cycle as a delay differential equation and solves it. Section 5 derives evolution as gradient ascent on a Fluidity landscape. Section 6 states the experimental predictions.

## 2 The Four States of a Redox System

Every electron-transfer system—a battery, a seed, an organism, a species—occupies one of four states defined by the electron flow $I$ across its boundaries and its Reinitiation Capacity $\Psi$ [6]:

Table 1: The four states of a redox system.

| **State** | $I$ | $\Psi$ | Example |
|---|---|---|---|
| Active | $> 0$ | $> 0$ | living organism; charging battery |
| Dormant | $= 0$ | $> 0$ | seed, hibernator, 0% SOC battery |
| Crystallized | $= 0$ | $= 0$ (latent) | offspring (zygote, stored RNA) |
| Dead | $= 0$ | $= 0$ (irreversible) | deceased organism; dead battery |

The borders between these states are the biology of life and the engineering of storage:

**Definition 1** (Reinitiation Capacity)**.** *The **Reinitiation Capacity** is the maximum free energy available to restore the system to its Active state:*

$$\Psi = \max(\Delta G_{\text{restore}}). \tag{4}$$

**Law 1** (Life Criterion)**.** *A system is* alive *iff* $\Psi > 0$. *A dormant or crystallized system has not lost life; it has merely stopped flowing.*

**Law 2** (Entropic Death)**.** *The Reinitiation Capacity decays toward zero:*

$$\frac{d\Psi}{dt} = -k_{\text{entropic}}\,\Psi, \tag{5}$$

*and* death occurs when $\Psi = 0$.

This is the same physics as a battery: dormancy is a living 0% state of charge; death is irreversible loss of the restore gradient. Aging is the progressive decay of $\Psi$; caloric restriction, hormesis, and repair are, in TRT language, down-regulations of $k_{\text{entropic}}$ [4].

## 3 Reproduction as a Triadic Cycle

### 3.1 The Triadic Nature of Sex

Sex is electron transfer between a donor and an acceptor across a mediating relationship:

- **Male (H-like):** donor, disperser, innovator, risk-taker.
- **Female (He-like):** acceptor, nurturer, stabilizer, conservator.
- **The relationship (Middle Fluid):** oscillator between H and He states.

The hormonal triad maps exactly onto the triadic roles:

Table 2: The hormonal triad.

| **Hormone** | **TRT role** | **Behavioral manifestation** |
|---|---|---|
| Testosterone (T) | H-like | risk-taking, competition, novelty-seeking |
| Estrogen (E) | He-like | nurturing, bonding, stability |
| Oxytocin (OXT) | Middle Fluid | empathy, attachment, emotional flexibility |

A relationship, like a battery electrolyte, has a healthy window:

$$0.5 < \Phi_{\text{relationship}} < 2.0, \qquad \Phi_{\text{relationship}} \approx 1.6 \text{ healthy}, \tag{6}$$

with the two classic failure modes being the two classic battery failure modes:

- **Codependency,** $\Phi \to 0$: the Fluid freezes; no electron flow, no adaptation.
- **Chaos,** $\Phi \to \infty$: the Fluid is uncontrolled; metallic short-circuit, no structure.

### 3.2 The Offspring as Crystallized Fluid

The offspring is not a third parent. It is the *crystallized record of the parents' oscillation at conception* [8]:

$$\Phi_{\text{offspring}}(0) = \frac{\Phi_{\text{father}}(t_c) + \Phi_{\text{mother}}(t_c)}{2} + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2). \tag{7}$$

The zygote is in the **Crystallized** state: $\Psi$ latent, maximal store, zero flow. Development is then a pattern of re-liquefaction, formalized for puberty [7, 8]:

$$\Phi_{\text{puberty}}(t) = \Phi_{\min} + (\Phi_{\max} - \Phi_{\min}) \left(1 - e^{-k_{\text{puberty}}(t - \tau_{\text{puberty}})}\right) + S\, \delta_{\text{sex}}. \tag{8}$$

The complete life cycle is the **Transgenerational Triadic Cycle**:

*H-like (Father) + He-like (Mother) → Middle Fluid (Conception) → Crystallized Fluid (Offspring) → Re-liquefaction (Puberty) → New H-like (Son) or He-like (Daughter).*

Within a generation, Fluidity follows the OFAG oscillation:

$$\text{conception (reset)} \to \text{gestation (crystallization)} \to \text{birth (dormancy)} \to \text{puberty (re-liquefaction)} \to \text{adulthood (ac} \tag{9}$$

# 4 The Transgenerational Triadic Cycle (TTC)

## 4.1 The delay differential equation

Formally, the Fluidity of an individual in generation $g$ evolves within a generation as

$$\frac{d\Phi_g(t)}{dt} = -\lambda\Phi_g(t) + \mu \sin\left(\frac{2\pi t}{\tau_{\text{dev}}}\right) + \nu\Phi_{\text{env}}(t) + \xi(t), \tag{10}$$

where $\lambda$ is the aging/decay constant (entropy production rate), $\mu$ and $\tau_{\text{dev}}$ are the development oscillation amplitude and period, $\nu$ is the environmental sensitivity, and $\xi(t)$ is noise.

Combining Eq. (10) with the conception reset of Eq. (7) yields the *delay differential equation* for the lineage Fluidity at a fixed developmental age:

$$\frac{d\Phi(t)}{dt} = -\lambda\Phi(t) + \beta\Phi(t-\tau) + \nu\Phi_{\text{env}}(t) + \mu \sin\left(\frac{2\pi t}{\tau_{\text{dev}}}\right) + \xi(t), \tag{11}$$

with the *heritability coefficient* $\beta$ (fraction of parental Fluidity transmitted) and *generation time* $\tau$.

## 4.2 Steady state and the critical heritability threshold

For constant environment $\Phi_{\text{env}} = \Phi_0$, the steady-state lineage Fluidity is

$$\Phi_{\text{steady}} = \frac{\nu\Phi_0}{\lambda - \beta e^{-\lambda\tau}}. \tag{12}$$

The steady state exists and is positive iff $\beta e^{-\lambda\tau} < \lambda$, which defines the **critical heritability threshold**:

**Theorem 1** (Stability / critical heritability)**.** *The steady state of Eq. (11) is asymptotically stable iff $\beta e^{-\lambda\tau} < \lambda$, i.e.*

$$\beta < \beta_{\text{crit}} = \lambda e^{\lambda\tau}. \tag{13}$$

*Proof.* Seek solutions $\Phi(t) = e^{st}$; the characteristic equation is $s = -\lambda + \beta e^{-s\tau}$. At the stability boundary take $s = i\omega$; separating real and imaginary parts gives $\cos\omega\tau = \lambda/\beta$ and $\omega = -\beta\sin\omega\tau$, whence $\sin\omega\tau = \sqrt{1-(\lambda/\beta)^2}$ and the second equation requires $\omega\tau = \arccos(\lambda/\beta)$ with $\beta e^{-\lambda\tau} = \lambda$. For $\beta e^{-\lambda\tau} < \lambda$ all roots have $\text{Re}(s) < 0$ and the lineage is stable. □ □

The interpretation is sharp [9]:

- $\beta < \beta_{\text{crit}}$: the lineage approaches a healthy, stable steady state.
- $\beta > \beta_{\text{crit}}$: the lineage diverges—hyperactive chaos ($\Phi \to \infty$) or entropic death ($\Phi \to 0$).

- $\beta = \beta_{\text{crit}}$: bifurcation; small perturbations drive extinction.

For the numerical human benchmark ($\lambda = 0.05$ yr$^{-1}$, $\tau = 20$ yr):

$$\beta_{\text{crit}} = 0.05\, e^{1.0} \approx 0.136, \tag{14}$$

so long-term stability requires *heritability below 13.6%.* This matches the striking empirical fact that the heritability of lifespan in humans is $\approx 25\%$, bounded below 30% in twin studies—natural selection keeps $\beta$ near its critical boundary, which is exactly where multi-generational oscillation (boom–bust) and adaptive reserves are largest.

### 4.3 Transgenerational memory

A step change in environment ($\Phi_{\text{old}} \to \Phi_{\text{new}}$) relaxes the lineage exponentially with rate

$$\gamma = \lambda - \beta e^{-\lambda\tau}, \tag{15}$$

i.e. transgenerational memory has half-life

$$t_{1/2} = \frac{\ln 2}{\lambda - \beta e^{-\lambda\tau}}. \tag{16}$$

This is the mathematical basis for *transgenerational epigenetic inheritance*: nutritional or toxic challenges echo for generations, and persist longer in lineages with high $\beta$ and large $\tau$ [9]. For the benchmark lineage the environmental-shock half-life is $\approx 18$ yr—about one generation.

### 4.4 Oscillation, filtering, and optimal generation time

Near the stability boundary the lineage oscillates with generational frequency

$$\omega = \sqrt{\beta^2 - \lambda^2}, \tag{17}$$

producing boom–bust cycles in populations. Under sinusoidal environmental forcing the lineage acts as a *filter*: it amplifies oscillations near $\omega$ and attenuates high frequencies [9]. The steady Fluidity is maximized at the optimal generation time

$$\boxed{\tau_{\text{opt}} = \frac{1}{\lambda} \ln\left(\frac{\beta}{\lambda}\right)} \tag{18}$$

a quantitative statement of the classic life-history tradeoff: long generations maximize steady Fluidity at the price of slower feedback control, and short generations buy agility at the price of stability.

## 5 The Unified Triadic Theory of Evolution (UTTE)

### 5.1 The fitness landscape

The TTC gives evolution a physical definition of fitness [10]: the time-integrated probability that the lineage stays alive, i.e. inside the Operational Window,

$$W(\boldsymbol{\theta}) = \int_0^\infty \mathbb{P}(0.5 < \Phi_g(t) < 2.0)\, dt, \qquad \boldsymbol{\theta} = (\lambda, \beta, \tau, \nu, \mu, \sigma^2). \tag{19}$$

Natural selection is gradient ascent on $W$ [2]:

$$\dot{\boldsymbol{\theta}} = \mathbf{K} \cdot \nabla_{\boldsymbol{\theta}} W(\boldsymbol{\theta}) + \boldsymbol{\eta}(t), \tag{20}$$

where $\mathbf{K}$ is the additive genetic variance–covariance matrix (the $G$-matrix) and $\boldsymbol{\eta}(t)$ is stochastic noise (mutation and drift). Setting $\nabla_{\boldsymbol{\theta}} W = 0$ yields the optimal lineage parameters:

Table 3: Optimal lineage parameters under pure selection.

| **Parameter** | **Optimum** |
|---|---|
| Heritability $\beta$ | $\beta_{\text{opt}} = \lambda e^{\lambda\tau} \cdot \dfrac{\nu\Phi_{\text{env}}}{\lambda + \nu\Phi_{\text{env}}}$ |
| Generation time $\tau$ | $\tau_{\text{opt}} = \dfrac{1}{\lambda} \ln(\beta/\lambda)$ |
| Aging rate $\lambda$ | $\lambda_{\text{opt}} = \dfrac{\ln(\beta/\tau)}{\tau}$ |
| Plasticity $\nu$ | $\nu_{\text{opt}} = \dfrac{\lambda - \beta e^{-\lambda\tau}}{\Phi_{\text{env}}}$ |
| Mutation rate $\sigma^2$ | $\sigma^2_{\text{opt}} = \dfrac{\lambda\Phi_{\text{env}}}{\beta\tau} \cdot \dfrac{d\Phi_{\text{env}}/dt}{\Phi_{\text{env}}}$ |

### 5.2 The five laws of evolution

**Law 3** (Fluidity Maintenance)**.** *Lineages survive iff they maintain* $0.5 < \Phi(t) < 2.0$ *across generations. Extinction occurs when* $\Phi$ *crosses 0.5 (entropic death) or 2.0 (chaotic collapse).*

**Law 4** (Heritability Bounds)**.** $\beta < \beta_{\text{crit}} = \lambda e^{\lambda\tau}$ *is required for long-term stability, and evolution drives* $\beta \to \beta_{\text{opt}}$ *given in Table 3.*

**Law 5** (Optimal Generation Time)**.** *Generation time evolves toward* $\tau_{\text{opt}} = (1/\lambda)\ln(\beta/\lambda)$*: long generations in stable environments, short generations in changing ones.*

**Law 6** (Environmental Tracking)**.** *Plasticity evolves to match the rate of environmental change,* $\nu_{\text{opt}} = (\lambda - \beta e^{-\lambda\tau})/\Phi_{\text{env}}$*: high plasticity in unpredictable environments, low in predictable ones.*

**Law 7** (Adaptive Innovation)**.** *The mutation rate balances novelty against mutational load:* $\sigma^2_{\text{opt}} = (\lambda\Phi_{\text{env}}/\beta\tau) \cdot (\dot{\Phi}_{\text{env}}/\Phi_{\text{env}})$.

### 5.3 Environmental regimes and the master equation

The four canonical environments select four distinct life histories [10]:

- **Constant** (e.g. deep-sea vents): long-lived, low mutation, low plasticity.
- **Trending** (e.g. climate change): short generations, high mutation, high plasticity—the species that track a trend are the fluid ones.
- **Cyclic** (e.g. seasonal): plasticity matches the environmental amplitude, $\nu \approx \Phi_1/\Phi_0$, with synchronized reproduction.
- **Stochastic** (e.g. deserts): bet-hedging—low heritability, short generations, high plasticity, high mutation.

The whole theory compresses into one master equation,

$$\boxed{\frac{d}{dt}\begin{pmatrix}\lambda\\ \beta\\ \tau\\ \nu\\ \mu\\ \sigma^2\end{pmatrix} = \mathbf{K}\cdot\nabla_{\boldsymbol{\theta}} W(\boldsymbol{\theta}) + \boldsymbol{\eta}(t)} \tag{21}$$

which unifies Darwinian selection ($\nabla W$), Mendelian inheritance ($\mathbf{K}$), epigenetic transmission ($\beta$ includes epigenetic effects), mutation and drift ($\boldsymbol{\eta}$)[10].

## 6 Testable Predictions

**Hypothesis 1** (H$_1$: Heritability–lifespan bound)**.** *The heritability of lifespan is bounded by* $\beta_{\text{crit}} = \lambda e^{\lambda\tau}$ *and is* $< 30\%$ *in humans (observed* $\approx 25\%$*). Species with high heritability have shorter potential lifespans.*

**Hypothesis 2** (H$_2$: Transgenerational decay)**.** *The effect of a nutritional or toxic intervention decays across generations with half-life* $t_{1/2} = \ln 2/(\lambda - \beta e^{-\lambda\tau})$*; measurable in multi-generational animal studies.*

**Hypothesis 3** (H$_3$: Generation–time tradeoff)**.** *Species with longer generation times must evolve lower heritability to remain stable (inverse relationship between* $\beta$ *and* $\tau$ *across taxa).*

**Hypothesis 4** (H$_4$: Population filtering)**.** *Populations filter environmental oscillations, amplifying frequencies near* $\omega = \sqrt{\beta^2 - \lambda^2}$*; detectable as resonant structure in population time series.*

**Hypothesis 5** (H$_5$: Stasis and punctuation)**.** *In stable environments* $\beta \to \beta_{\text{opt}}$, $\tau \to \tau_{\text{opt}}$, $\sigma^2 \to 0$*: evolution freezes (punctuated equilibrium). Extinction risk is highest for lineages with* $\beta > \beta_{\text{crit}}$*, within* $\tau \ln 2/(\beta - \lambda)$ *generations.*

## 7 Conclusion

TRT dissolves the wall between chemistry and biology by reducing life, sex, reproduction, and evolution to the tracking of one number. Life is the maintenance of $0.5 < \Phi < 2.0$; death is the irreversible loss of $\Psi$; reproduction is the passage of a crystallized Fluid between a donor and an acceptor; and evolution is gradient ascent in Fluidity-space. The mathematics is not metaphorical: the critical heritability threshold $\lambda e^{\lambda\tau}$ explains why lifespan heritability is stuck near 25%, the delay equation explains generational memory and boom–bust cycles, and the master equation binds Darwin, Mendel, and epigenetics into one dynamical system *from the electron to the ecosystem.*

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